How do young children learn and remember?

T/Th 9:35-10:25am

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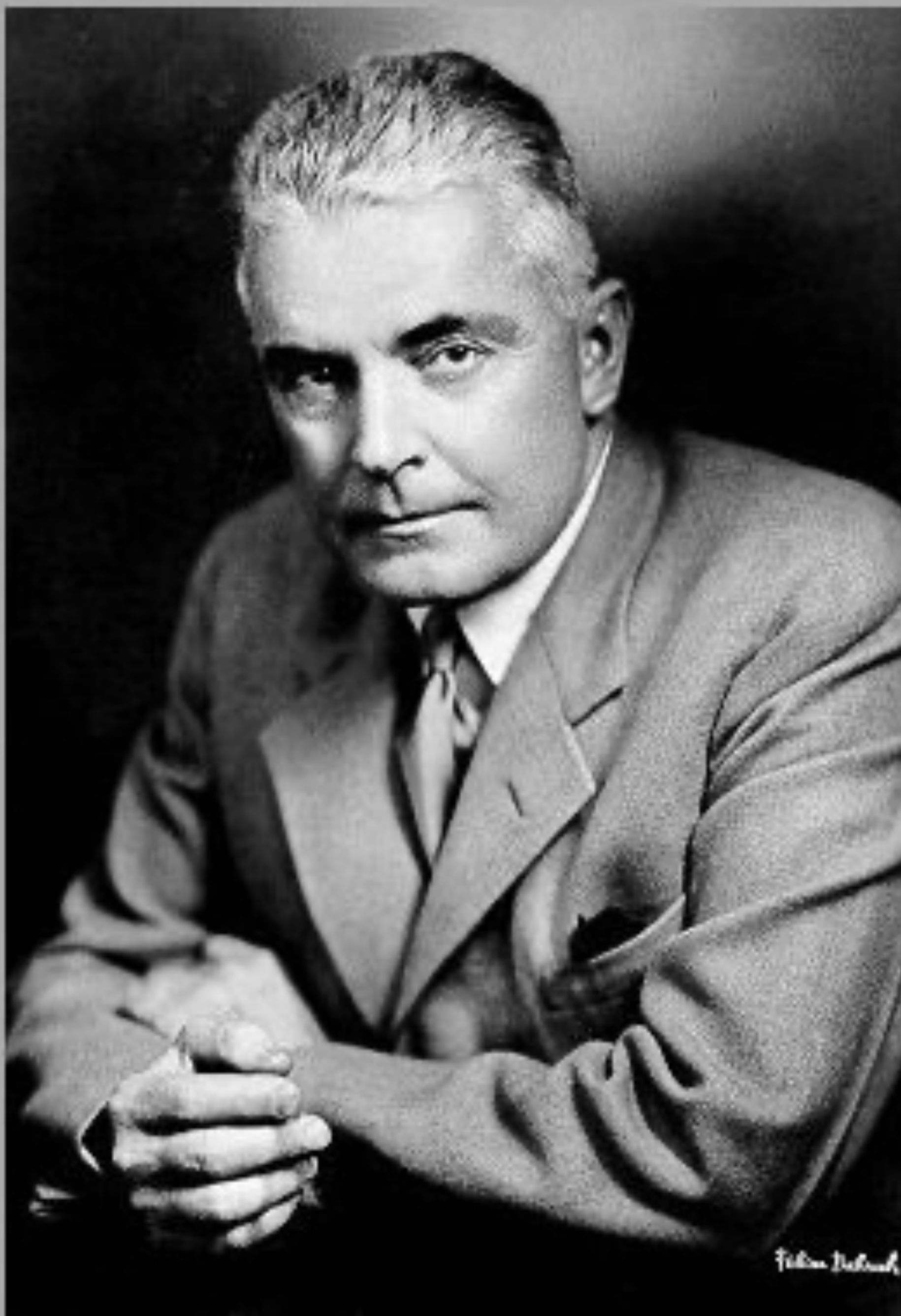
Today

- **Mechanisms of learning**
  - Classic learning theory: Behaviorism
  - Imitation
  - Social referencing
  - Prediction & prediction error
- **Memory**
  - What develops in memory?
  - Children’s eye-witness testimony

Classic learning theory

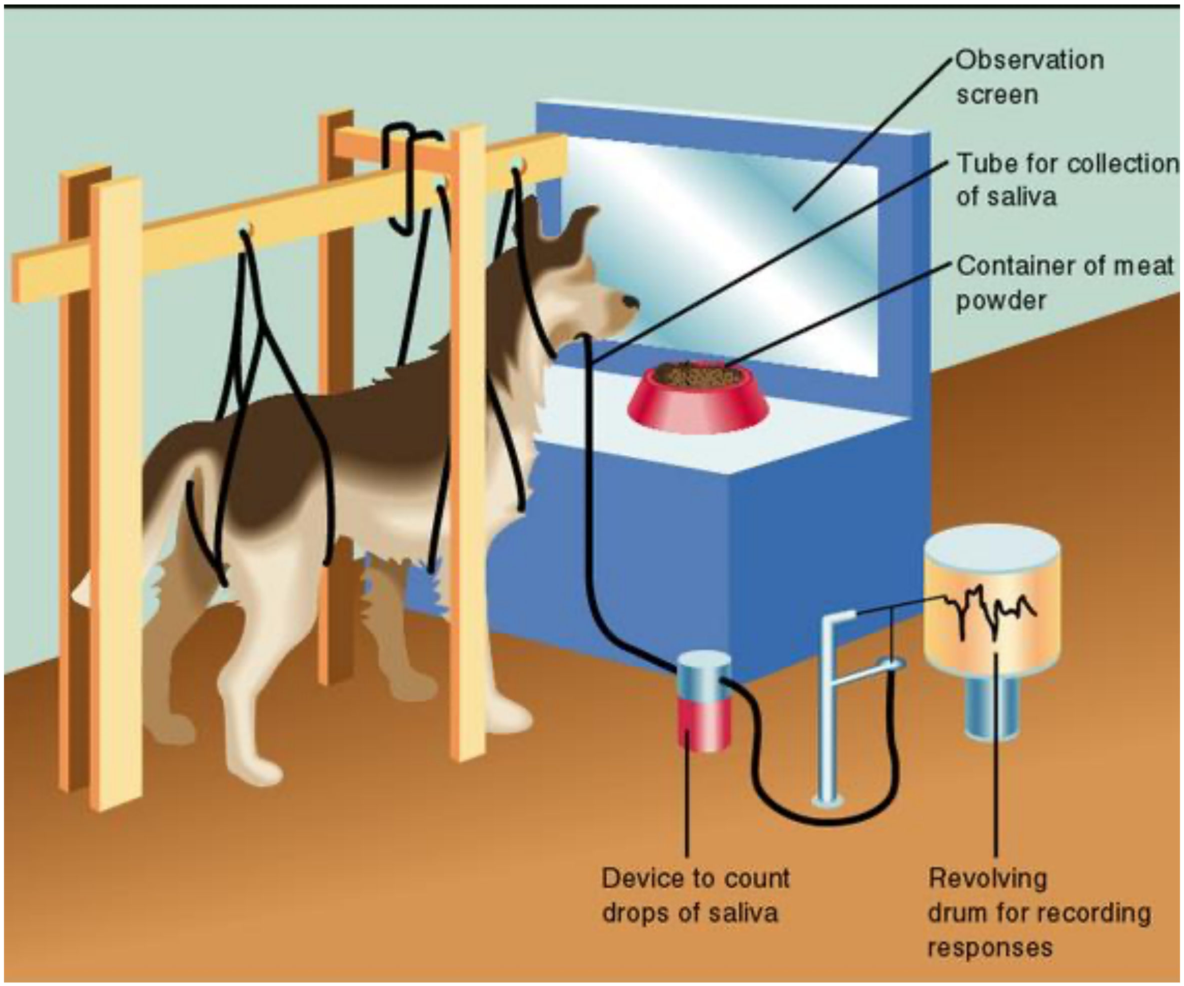
Classic learning theory: Behaviorism



"Give me a dozen healthy infants, well formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief, and yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors."

John Watson

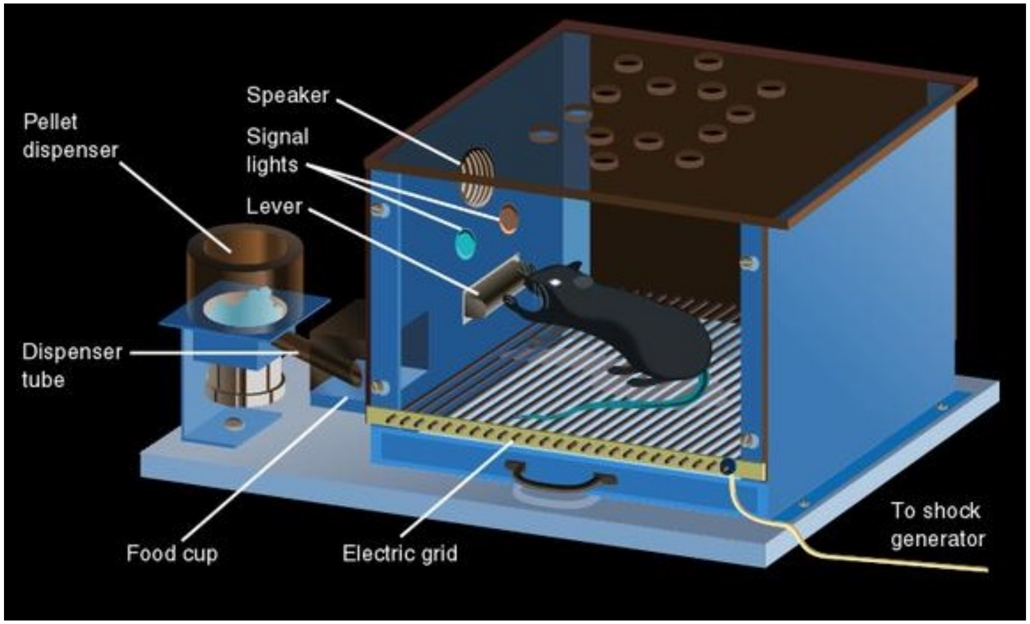
Classic learning theory: Pavlov's dogs



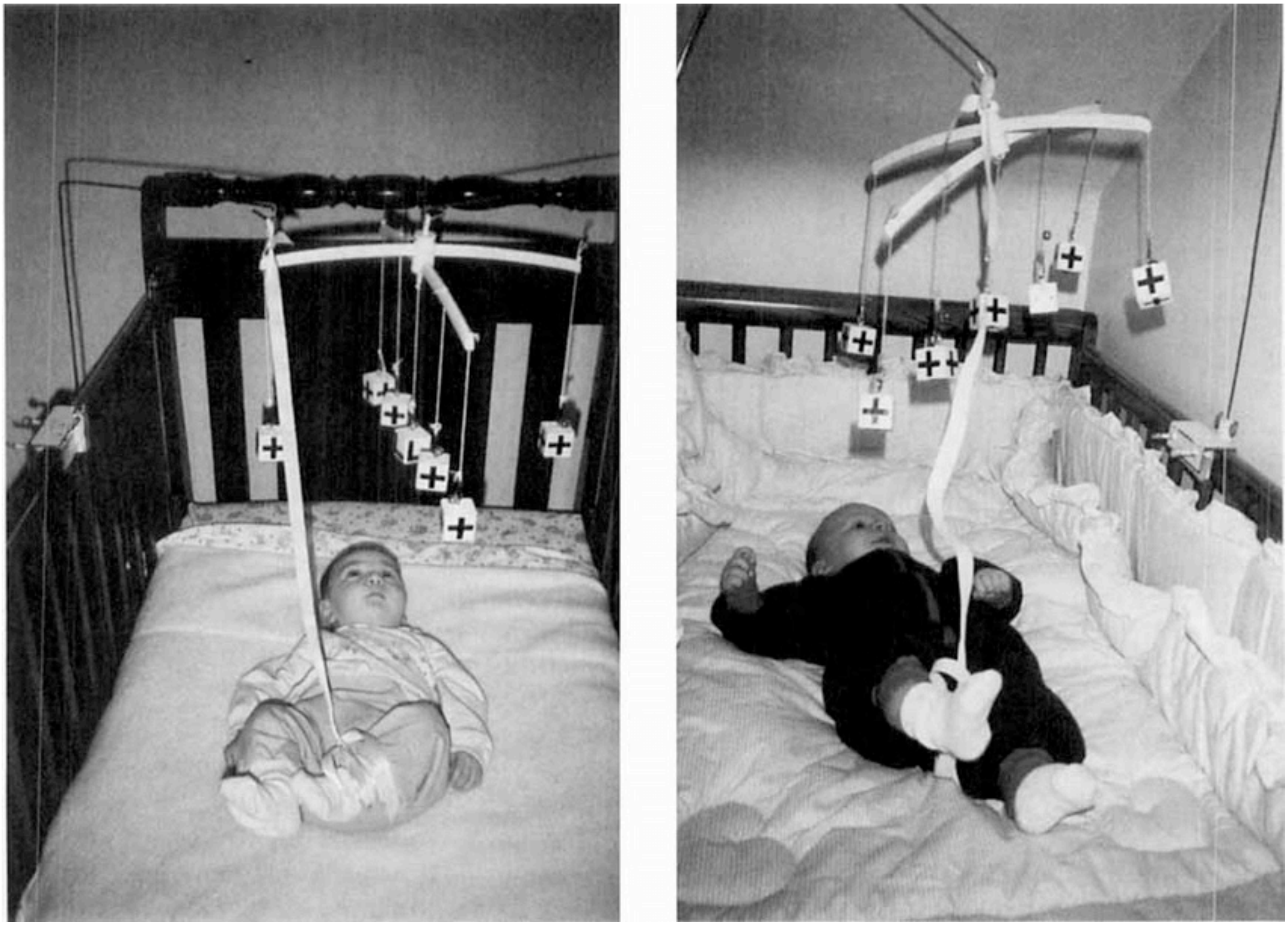
Classic learning theory: Pavlov’s dogs

Classic learning theory: Operant conditioning

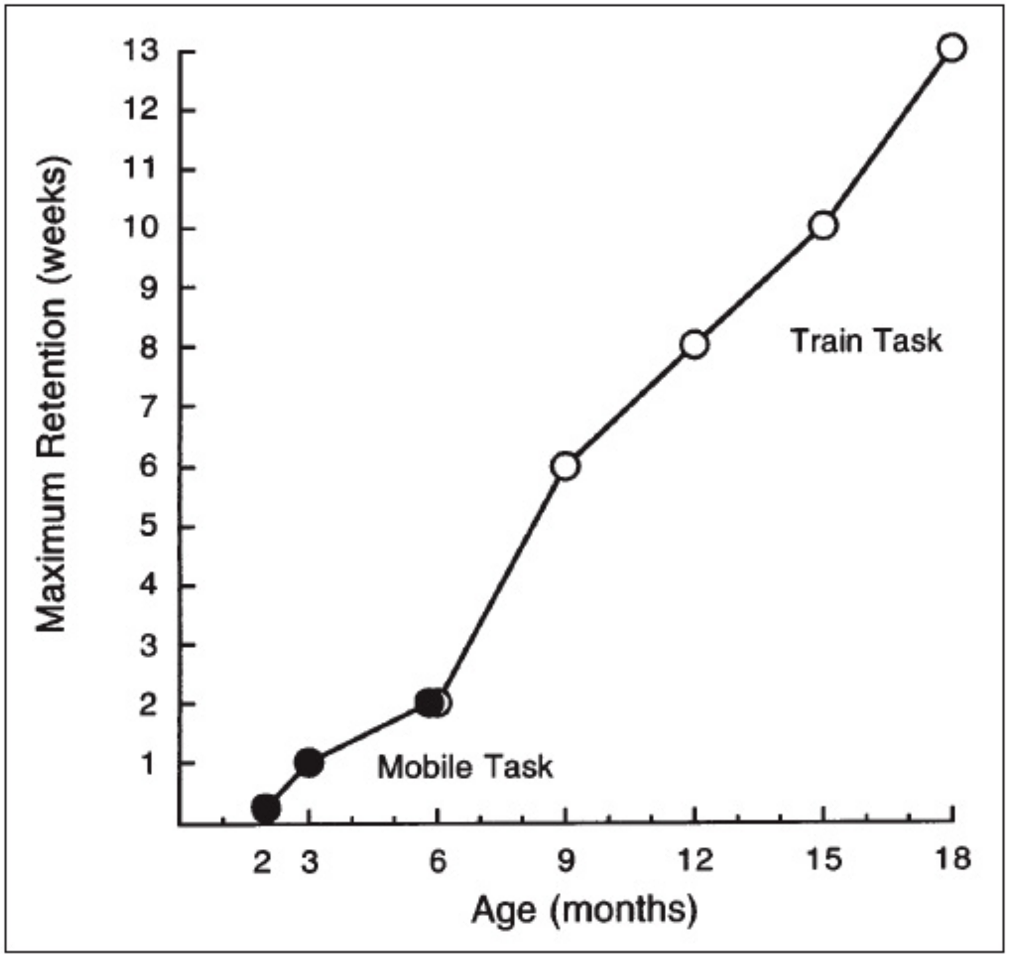
B.F. Skinner (1904-1990)



Example with infants: The mobile task



Example with infants: The mobile task



(But memory is context-dependent)

Imitation

Imitation

- Newborns (kind of) imitate!

(Meltzoff, 1989)

Imitation

Imitation of intentions

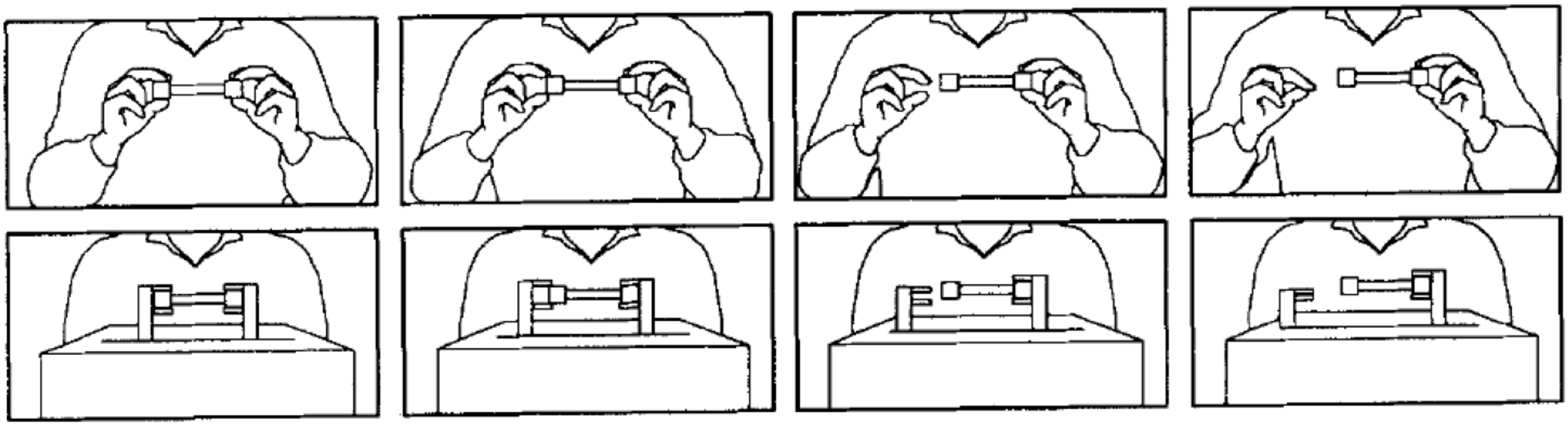
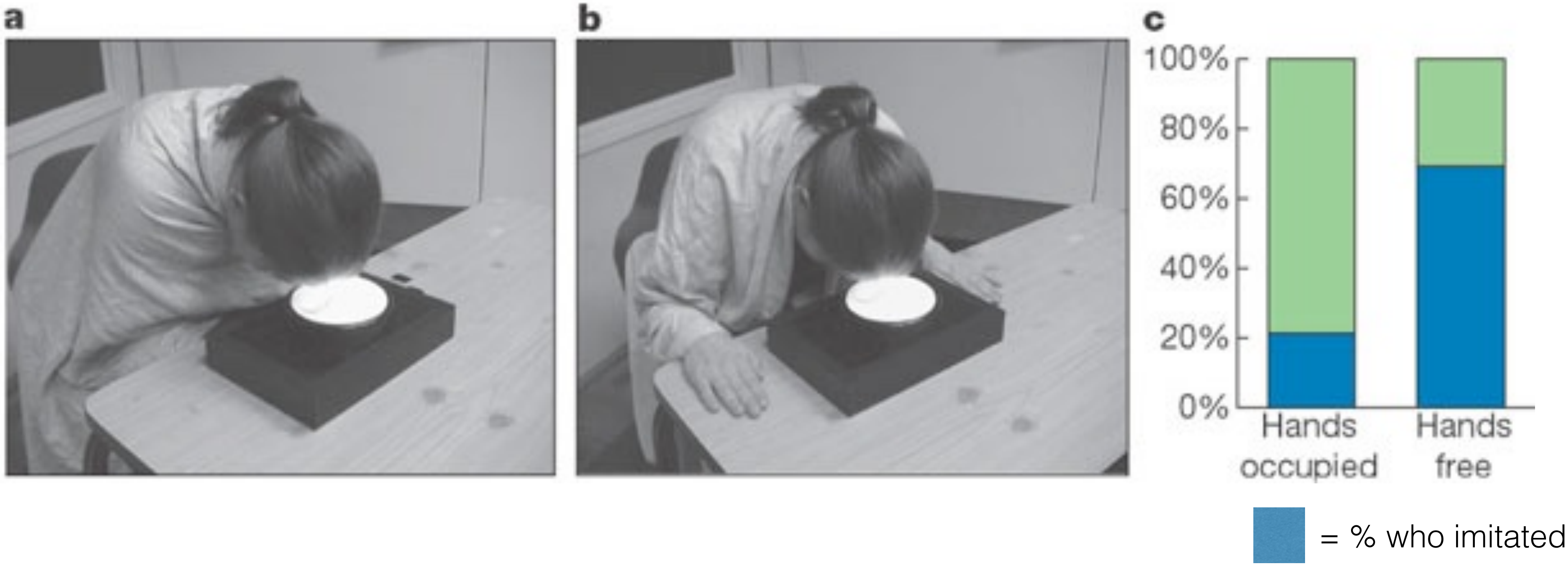


Figure 2. Human demonstrator (top panel) and mechanical device mimicking these movements (bottom panel) used in Experiment 2. Time is represented by successive frames left to right.

Rational imitation



(Buttelman et al., 2013)

Social referencing

Social referencing

Prediction & prediction error

Prediction error

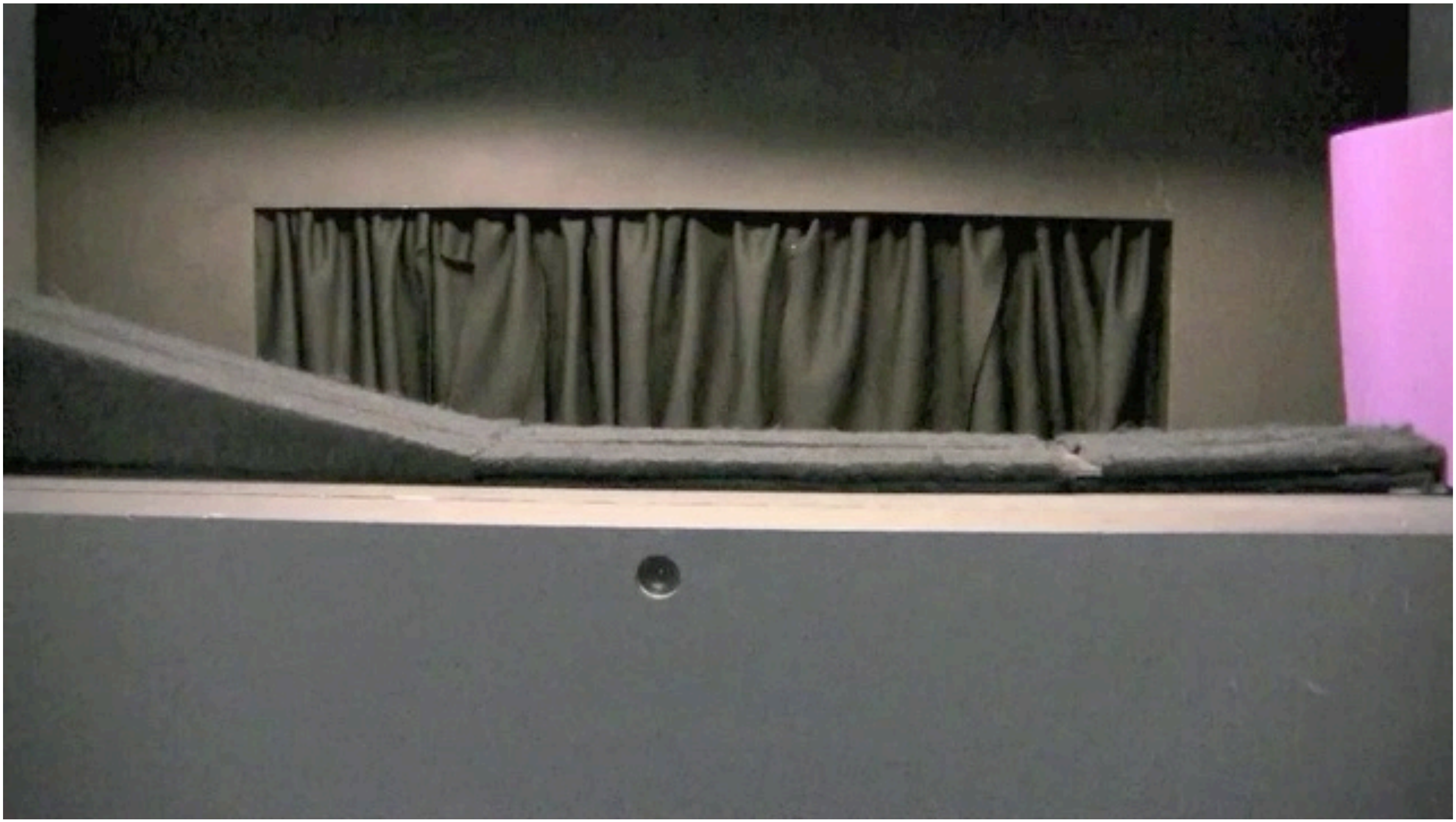
We track events from the past,
evaluate them in relation to the present environment,
and make predictions about the future.

Sometimes we're wrong.

Then what?

Violation of expectations

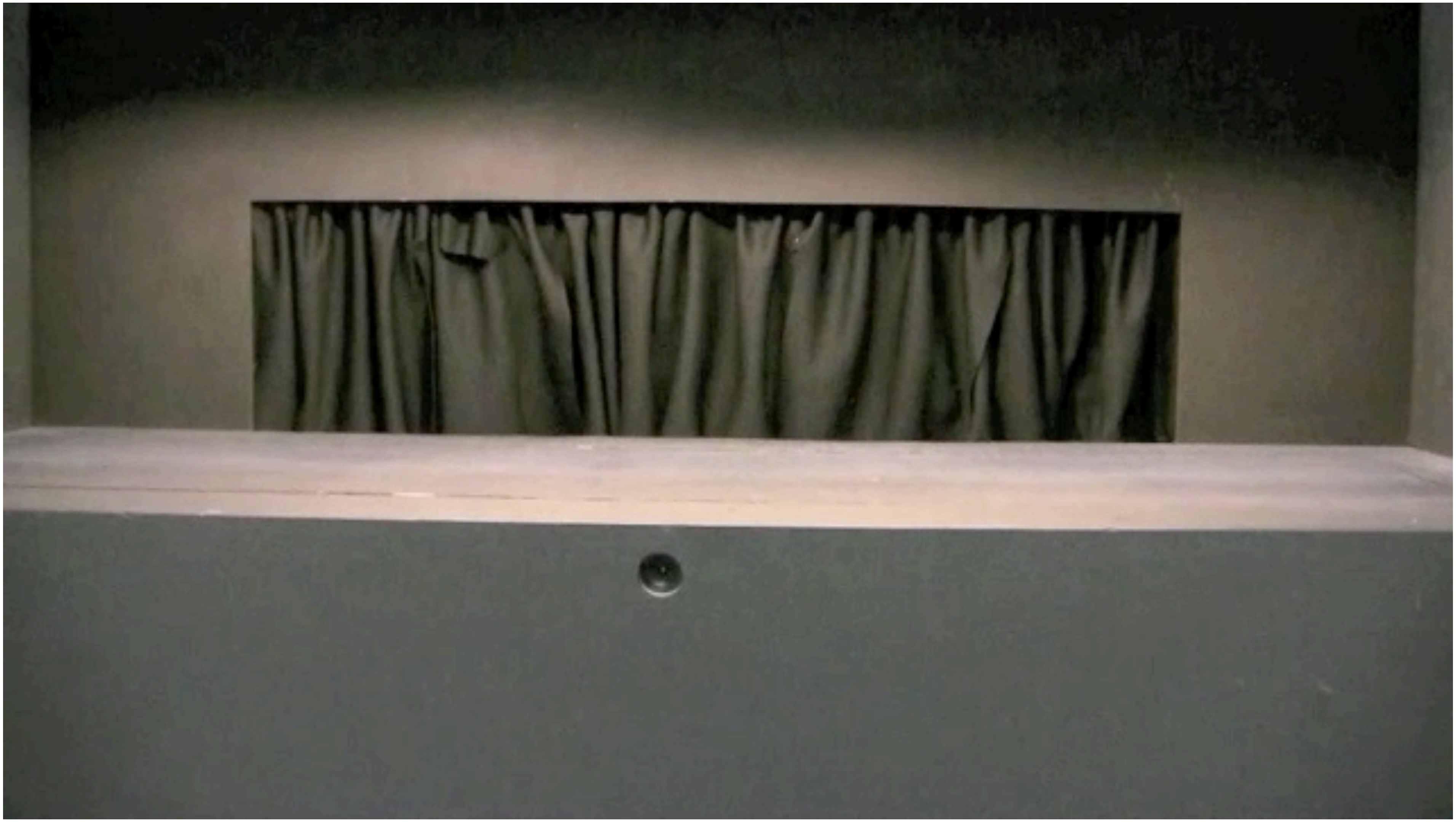
- Solidity violation



(Stahl & Feigenson, 2015)

Violation of expectations

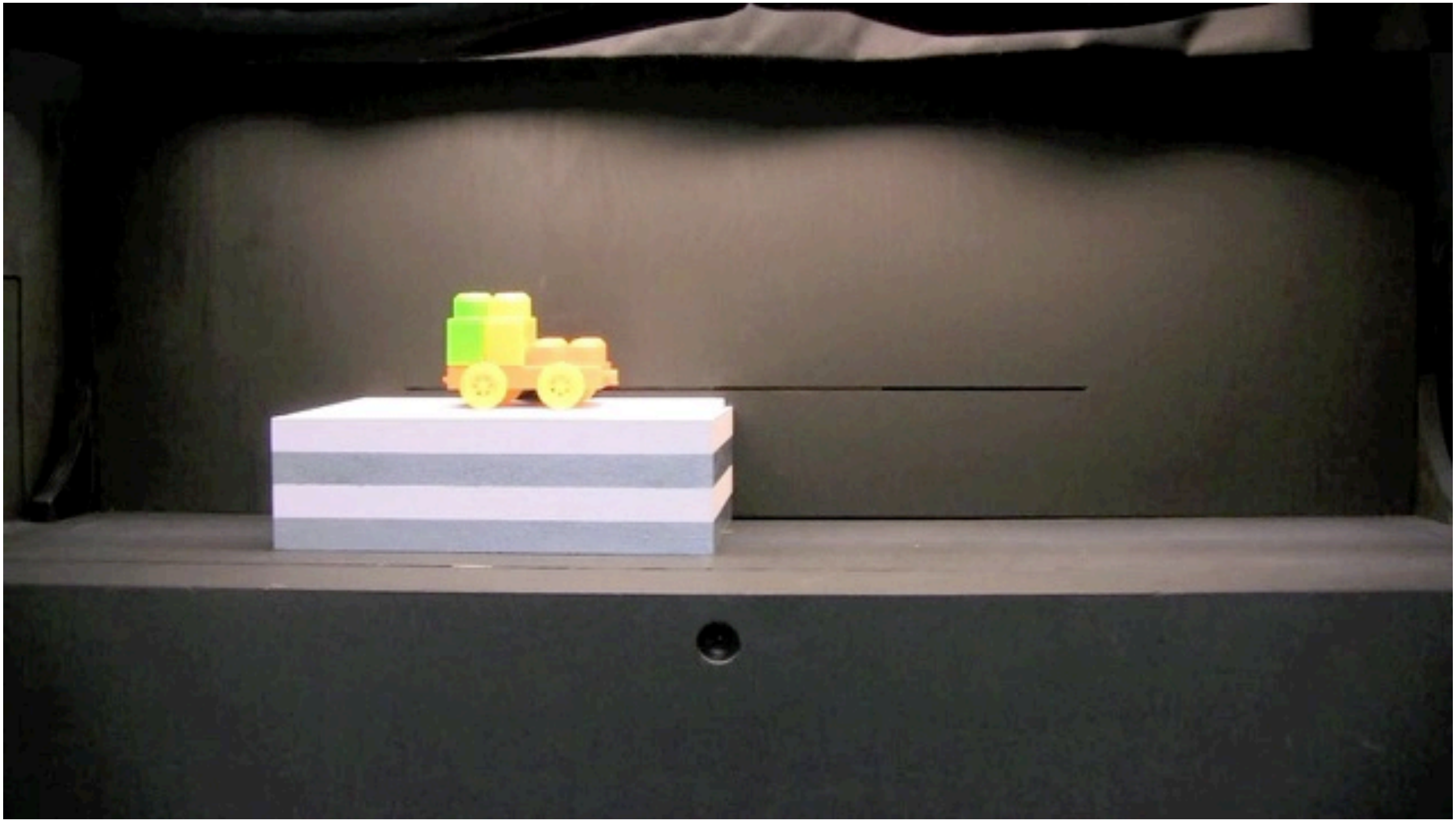
- Continuity violation



(Stahl & Feigenson, 2015)

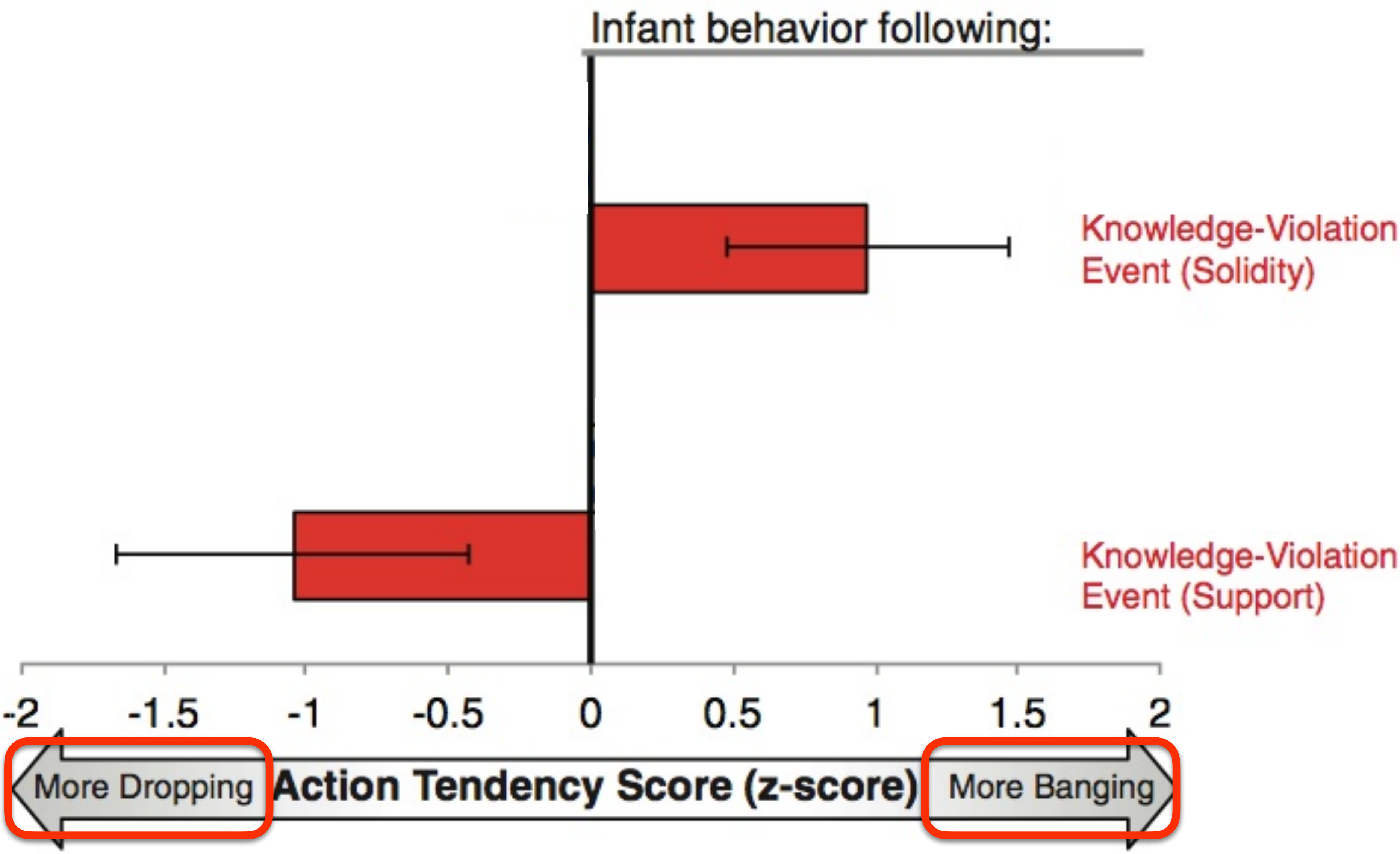
Violation of expectations

- Solidity vs. Support violation



(Stahl & Feigenson, 2015)

Violation of expectations



(Stahl & Feigenson, 2015)

Memory

We’ve already covered memory a little bit

- Object permanence; A-not-B
- Language (mom’s voice, Cat in the Hat, native language…)
- Habituation paradigm
- Remember the mobile task from 20 minutes ago?

Do infants remember?

- Infantile amnesia: adults' failure to remember early experiences
    - Most adults can't remember events from first 3-4 years
- But does this mean that infants don't form memories? Or that memories are encoded in a way that's hard to store and retrieve?
- Elaboration from caregivers

What develops in memory development?

1. **Basic processes**
2. **Strategies**
3. **Metacognition**
4. **Content knowledge**

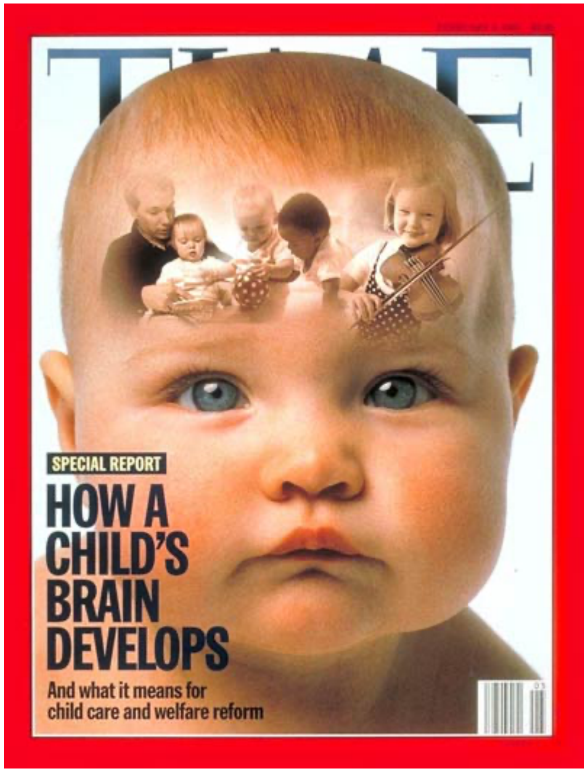
What develops in memory development?

- **Basic processes**

  - **Changes in brain structure**
    - The hippocampus and amygdala support memory formation, and they develop early
    - But frontal cortex is necessary to retrieve memories, and it develops later

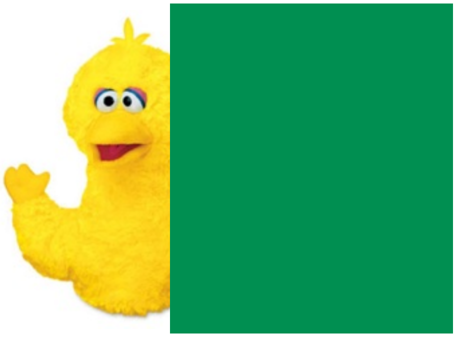
  - **Processing speed**

  - **Ability to generalize**



What develops in memory development?

- **Strategies**
    - Cognitive/behavioral activities under control of the subject, used to enhance memory
    - Early strategies are used inconsistently
    - Example: searching for objects

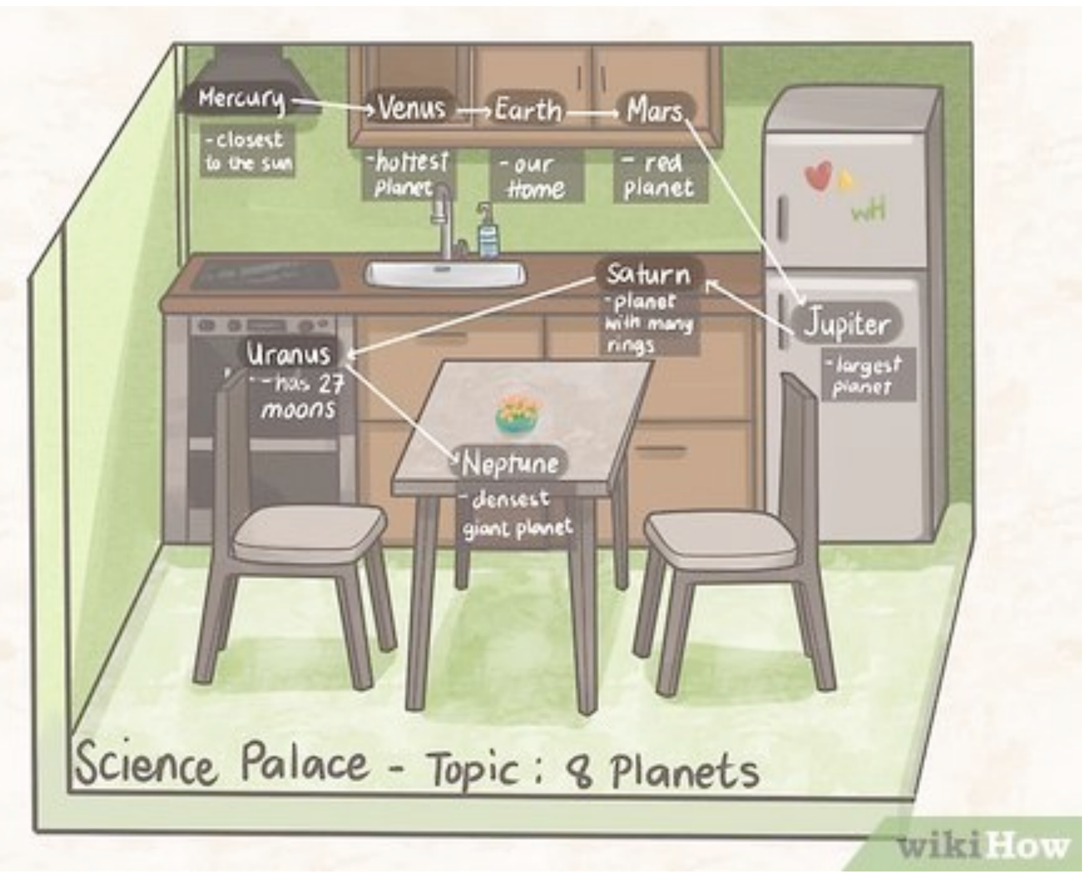


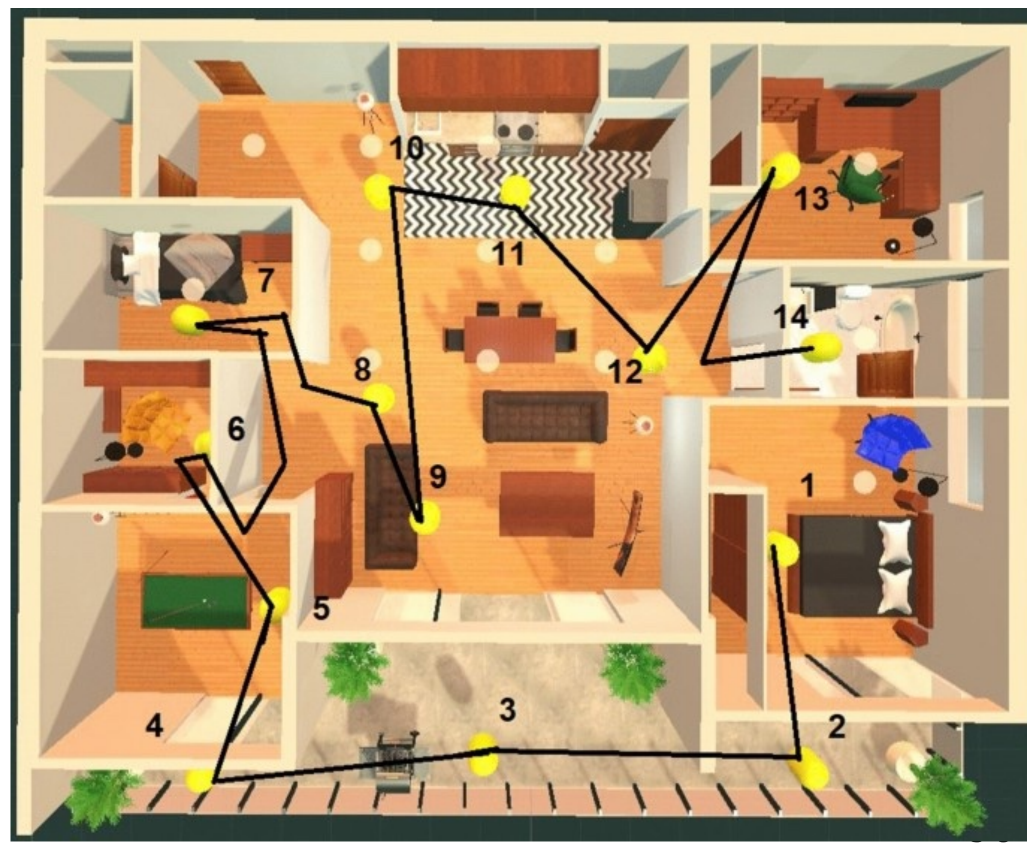
What develops in memory development?

- **Strategies**
  - **Rehearsal**: repeating information over and over

What develops in memory development?

- **Strategies**
  - **Elaboration**: creating meaningful relationships between items
    - My very excellent mother just served us nine (pizzas)
  - **Imagery**: method of loci, a.k.a. ‘memory palace’





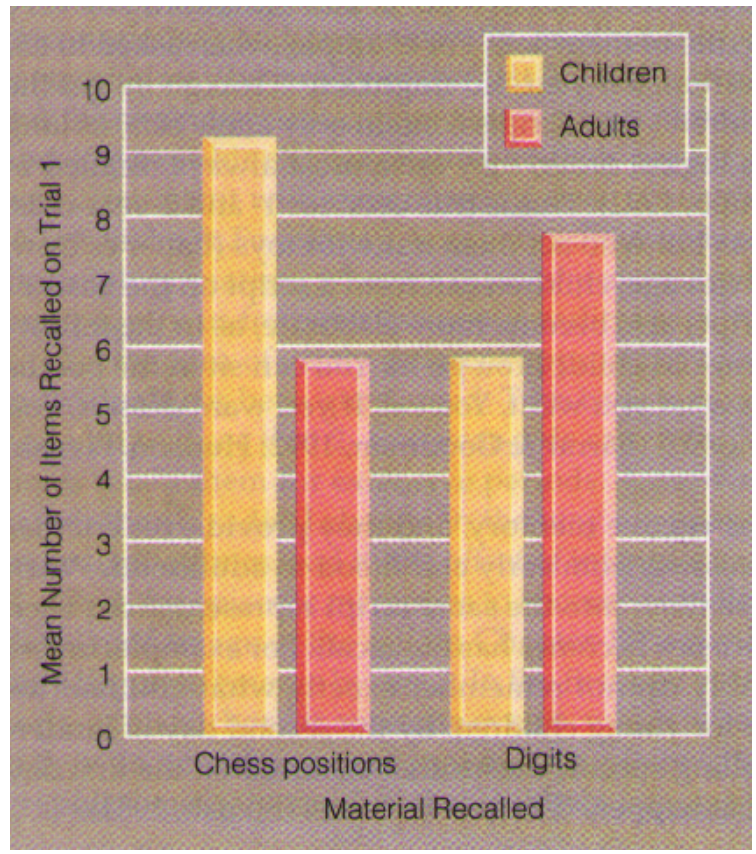
What develops in memory development?

- **Metacognition**
    - Knowledge/awareness of your own cognitive processes & control of cognitive processes
    - Requires self-monitoring (e.g., slowing down when reading a difficult sentence)
    - Can be an implicit or explicit process

What develops in memory development? 👀

- **Content knowledge**

    - Knowing more about the world helps memory
    - Children can have better memories than adults for subjects they are experts in



Children's eye-witness testimony

Children’s eye-witness testimony

Children's eye-witness testimony

- More than 100,000 children testify in legal cases each year
- Stakes are high
- **Are young children's memories reliable?**

- Characteristics of real forensic interviews that might influence children's memories:
  - Questions about personal, embarrassing, or stressful situations
  - Leading questions (influenced by interviewers' expectations)
  - Timeline: questioning weeks, months, years after the event

Children's eye-witness testimony

- Simon Says study
  - 3- to 6-year-old children played Simon Says, touching their own nose, stomach, foot, etc., and/or those of other children
  - One month later: social worker interviewed children
  - Result: leading questions, changing answers
  - Accuracy with/without leading questions

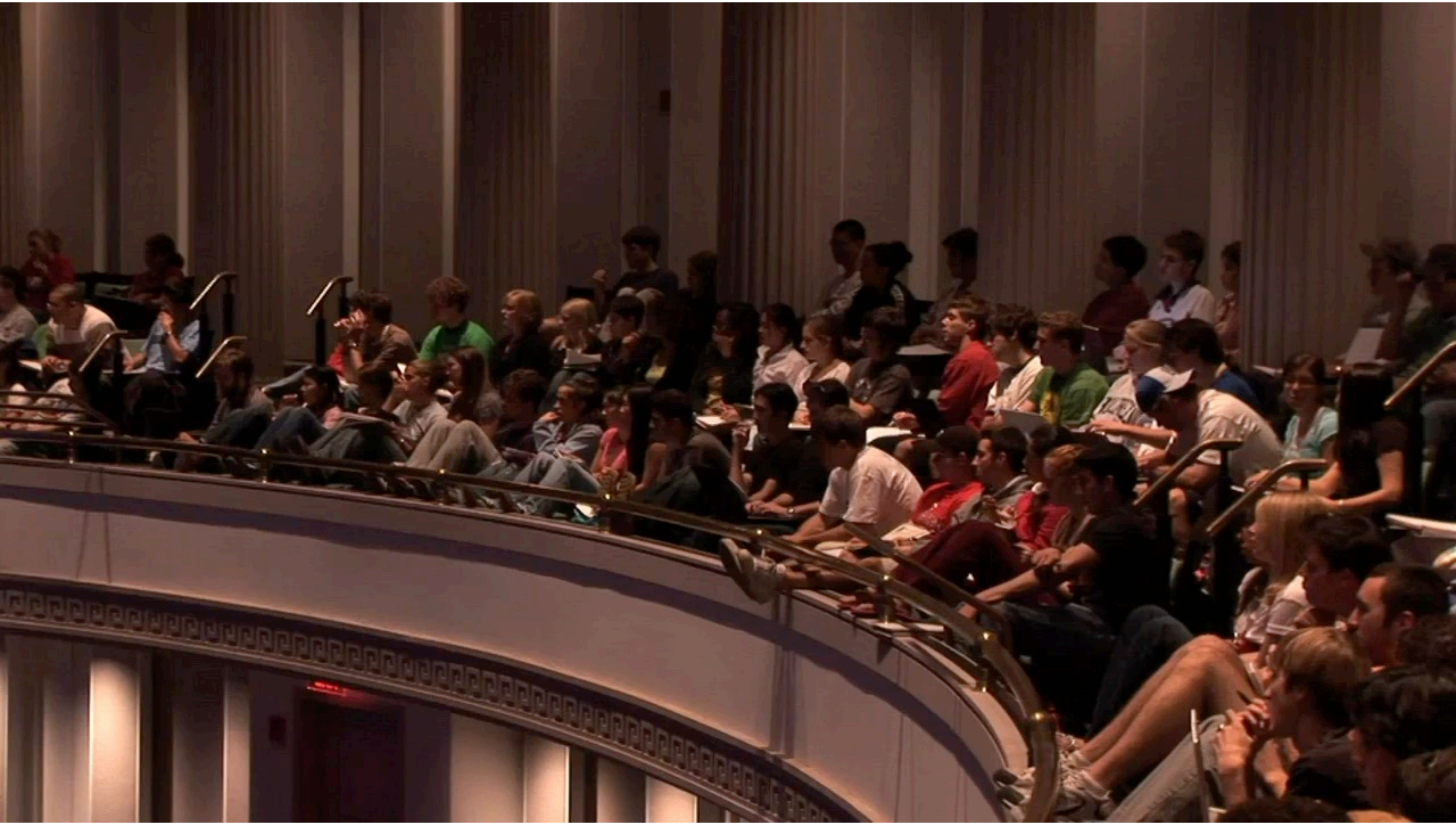
(Bruck et al., 2006; Ceci & Bruck, 1998; Poole, Bruck, & Pipe, 2011)

Children's eye-witness testimony

- Authority figure vs. civilian
- Realistic props
- Harmful for the child?
- Live vs. pre-recorded testimony
- Accuracy of children vs. adults

(Bruck et al., 2006; Ceci & Bruck, 1998; Poole, Bruck, & Pipe, 2011)

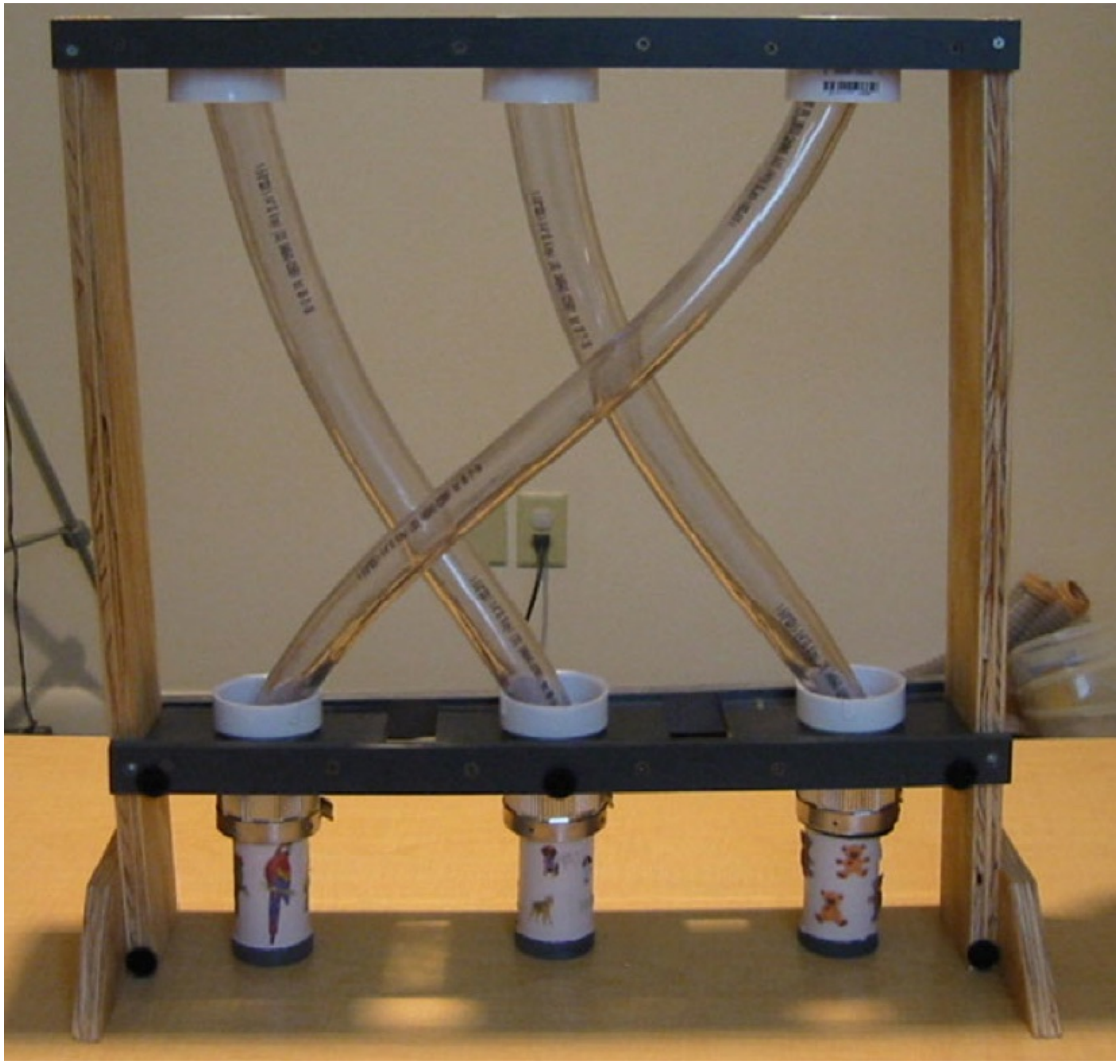
Children's eye-witness testimony



<u>Stereotypes</u>: Clumsy Sam study

Children's eye-witness testimony

- Why are some children consistently willing to believe what they're told, even when it conflicts with first-hand experience?



(Jaswal et al., 2014)

Children’s eye-witness testimony

- Are these studies applicable to real life?

(Bruck et al., 2006; Ceci & Bruck, 1998; Poole, Bruck, & Pipe, 2011)